

Salvatore T. Ciminello

(914) 268-7025 | stc92@cornell.edu | [Portfolio](#) | [linkedin.com/in/sal-ciminello](https://www.linkedin.com/in/sal-ciminello)

EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Mechanical Engineering, GPA: 3.71
Master of Engineering, Aerospace Engineering, GPA: 4.02

May 2026
Expected December 2026

Relevant Courses (* = M.Eng Graduate Level):

Dynamics and Controls: Intermediate Dynamics* | Mechatronics | System Dynamics | Autonomous Mobile Robotics*
Spacecraft Architecture: Propulsion of Aircraft & Rockets | Space Structures* | Spacecraft Systems Architecture*

Teaching Experience: System Dynamics (Lab Asst.) | Mechatronics (TA) | Dynamics (Grader)

TECHNICAL SKILLS

Hardware: Machining (Mill and Lathe), Rapid Prototyping, Additive Manufacturing, Electronics and Sensor Wiring, Feedback Control Systems

Software: MATLAB, Python, C++, Arduino, Raspberry Pi

CAD / Simulation: SolidWorks, Fusion 360, ANSYS Mechanical, ANSYS Maxwell, ANSYS Fluent

TECHNICAL EXPERIENCE

Cornell Aerospace Adversary Lab – Research Assistant, M.Eng Technical Project **January 2026-Present**
Air Force Research Laboratory (AFRL) funded | Prof. Falco (cybersecurity in space)

- Building a physical CubeSat constellation testbed to study cyber-attack propagation across distributed satellite networks, simulating how injected viruses spread under variable formation configurations.
- Designing and prototyping CubeSat structural mockups with intersatellite communication links; implementing distributed formation control to coordinate multi-satellite positioning during network-level attack scenarios.

Active and Passive Pod Control System – Hyperloop Project Team **August 2024-March 2026**

- Derived passive roll stability system by modeling pod as a spring-damper about its neutral axis via momentum balance; selected shock absorber and caster wheel configuration by calculating required damping coefficients analytically.
- Developed 1–4 DOF MATLAB simulations of active electromagnetic levitation, targeting a 4–5 mm air gap; characterized electromagnet force-distance behavior through physical testing and fit an exponential model to experimental data.

Autonomous Competition Robot – Mechatronics, Cornell University **January 2025-May 2025**

- Designed robot exoskeleton in SolidWorks with servo-driven chassis and block collection system; programmed Arduino integrating QTI, color, and proximity sensors for real-time navigation.
- Placed 5th out of 65 teams in the ASML–Cornell Robotics Competition.

Autonomous Mobile Robot – Autonomous Mobile Robotics, Cornell University **January 2026 - May 2026**

- Implemented robot localization using Kalman, Extended Kalman, and Particle Filters; developed RRT-based path planning for competitive autonomous navigation challenges.

Go2 – Mechanical Engineer & Co-Founder **May 2025-Present**

- Co-founded startup to develop a personal portable carbonation device; produced machine drawings and designed a pressure vessel with intake valve actuation and automatic pressure-relief system; coordinated overseas manufacturing to deliver first physical prototype.

LEADERSHIP EXPERIENCE

Mechanical Team Lead – Cornell Hyperloop Project Team **May 2025-Present**

- Led ~20 engineers across Structures, Magnetics, and Braking subteams; coordinated closely with a 20-person Electrical team and attended weekly cross-functional leads meetings spanning mechanical, electrical, and business.

Magnetics & Propulsion Subteam Lead – Cornell Hyperloop Project Team **December 2024-May 2025**

- Directed levitation, controls, propulsion, and cooling projects; designed multi-axis PID levitation control system using 4 proximity sensors and PWM controllers to regulate height and attitude across all 3 axes.